



# Saumitra Mazumder



SAMazumder



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## About Me

Quantitative finance and model-risk professional with experience spanning credit, market, liquidity, fraud, AML, and climate risk. Combines quantitative modeling and model validation expertise with hands-on development of Generative AI, LLM, RAG, agentic AI, and AI-assisted automation solutions for financial-services risk and regulatory workflows.

Experienced in developing and validating quantitative models, designing climate-adjusted risk methodologies, and applying machine learning to financial-risk problems. Currently focused on building AI-enabled tools and workflows that augment and automate model development, validation, analysis, and regulatory processes.

## Education

M.Sc. Applied Mathematics  
Toronto Metropolitan University  
2020-2022

B.Sc. Mathematics and Economics  
Toronto Metropolitan University  
2015-2019

## Technical Expertise

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|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI & Machine Learning | LLMs, Generative AI, RAG, agentic workflows, AI agents, embeddings, fine-tuning, model evaluation, prompt engineering, XGBoost, ML-based fraud/scam models, model evaluation, calibration, generalized additive models, time-series analysis.                                                                        |
| Quantitative Finance  | Credit risk, market risk, liquidity risk, VaR, CVaR, Expected Shortfall, stress testing, Monte Carlo methods, derivatives pricing, copulas, portfolio risk, IFRS 9 / CECL, model validation, conceptual soundness, data quality, calibration, performance monitoring, valuation models, regulatory model governance. |
| Tools                 | Python, OpenAI API, MCP/tool servers, TypeScript, C++, Java, SQL, R, SAS, MATLAB.                                                                                                                                                                                                                                    |

## Recent Employment History

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| 2026-Current | <p>PwC, LLP - Manager</p> <p>Financial Services Risk and Regulatory Consultant</p> <p>Development</p> <p>Lead developer of GenAI applications for Financial Services Risk &amp; Regulatory, including LLM-powered workflows, agent/tool integrations, RAG, and model-evaluation frameworks.</p> <p>Designed and implemented AI-assisted workflows to automate portions of quantitative model development, validation, documentation, and regulatory analysis.</p> <p>Built production-oriented AI tooling using Python, TypeScript, C++, Java, MCP/tool servers, and OpenAI APIs.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2023-2026    | <p>PwC, LLP - Senior Associate</p> <p>Corporate Financial Risk Management Advisory</p> <p>Development</p> <p>Led the development of implementing GenAI into a semi-automated model validation process. Experienced in building quantitative models across credit and market risk, as well as design and implementation of climate risk models (credit risk and market risk) under IPCC RCP/SSP scenarios, with heat map generation (physical risk and transition risk) and integration of climate-adjusted credit and market risk parameters into default risk frameworks.</p> <p>Development lead experience includes: (i) BRR/FRR models, (ii) Machine learning based scam &amp; fraud models, (iii) Rules based AML models, (iv) Credit risk models, including low default portfolios, as well as both retail &amp; wholesale portfolios, (v) Scenario-based analysis supporting valuation of complex exposures and stress testing under adverse market and credit conditions.</p> <p>Validation</p> <p>Extensive experience leading validation of quantitative models in compliance with regulatory and internal governance standards. Validation work spans credit, market, and liquidity risk models, including independent assessment of conceptual soundness, data quality, calibration, and performance monitoring.</p> <p>Validation lead experience includes: (i) IFRS 9/CECL ECL model validation for wholesale portfolios (Commercial, Corporate/Sovereign, and CRE), (ii) Valuation model validation, ensuring fair value estimation methodologies align with observable market inputs, (iii) Liquidity and stress testing frameworks, including cash flow forecasting, funding concentration risk, and market downturn stress scenarios, (iv) Fraud and AML model validation, assessing machine learning conceptual soundness under limited model development documentation and ensuring robustness and interpretability.</p> |